MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-UP, STEP-DOWN

The complete requirements for procuring the transformers herein shall consist of this document and the issue in effect of MIL-T-83720.

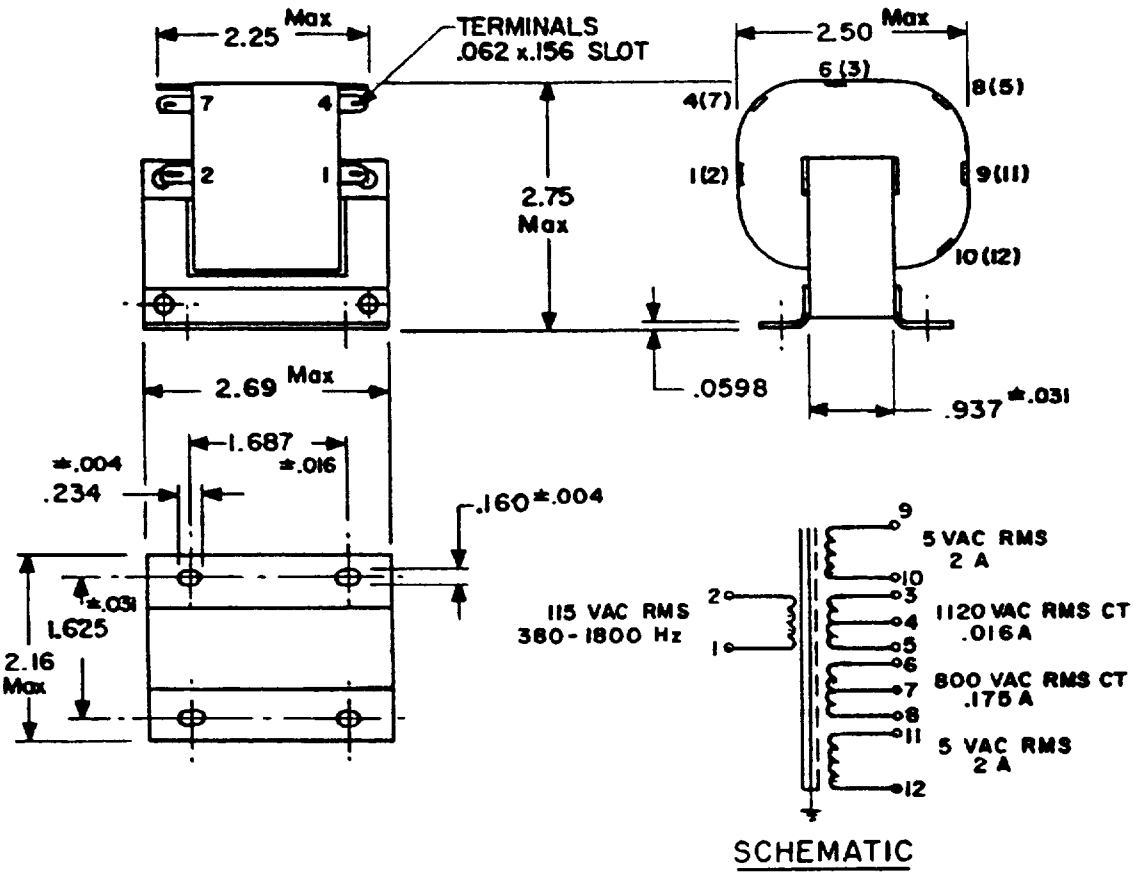


Figure 1

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:15Z
Check the source to verify that this is the current version before use.

REQUIREMENTS.

Mechanical and Environmental Requirements.

Dimensions and Configuration:	See Figure 1.
Vibration:	Test Condition A, Method 204.
Shock:	Test Condition C, Method 205.
Altitude:	10,000 feet.
Ambient Temperature Range:	-54°C to 85°C.
Maximum Operating Temperature:	170°C.
Temperature Rise:	80° maximum.
Weight:	1.12 nominal.
Terminal:	Solder type.

Electrical Ratings.

Working Voltage:	Primary 350 VAC rms; Secondary 1400 VAC rms.
Electrostatic Shield:	Min. Voltage Ratio 5:1 at 20 Hz.
Life:	10,000 hours.
Polarity:	Dot on schematic indicates positive sense.
Rating:	190 VA maximum for temperature rise of 80°C.
Duty:	Continuous.

Primary.

Phase:	Single.
Voltage:	115 V rms.
Frequency:	400-1800 Hz ± 20 Hz.

Other Electrical Ratings: See Table 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:15Z
Check the source to verify that this is the current version before use.

TABLE I.

Winding	Full Load Secondary Volts RMS at 115V ac Input	Rated Secondary Amps RMS at Full Load	Dielectric Test Voltage (RMS) at Sea Level
(1-2)			1500
(3-5)	1120 CT ± 4%	.016	3000
(6-8)	800 CT ± 4%	.12-	2600
(9-10)	5V ± 2%	2.0	3000
(11-12)	5V ± 2%	2.0	2600
Shield to windings			3000

NOTE: When numbers are used in parenthesis, e.g. (1-2), they indicate the winding and the extreme terminals of the winding.

NOTE: Winding (3-5), (6-8) are part of a full wave rectifier circuit with 50 Hy and 5 Hy inductive input filters respectively. Windings (9-10), (11-12) supply filament voltage to 5R4G7 rectifier tubes.

Part Number: Consists of the basic number of this specification sheet and a dash number.

Example: Part Number. M83720/13-01

Custodian:
Air Force - 85

Review Activities:
Air Force - 11, 84

Preparing Activity:
Air Force - 85

(Project 5950-F089-013)

Source: https://assist.dla.mil -- Downloaded: 202610-23T09:15Z
Check the source to verify that this is the current version before use.